

Programming Fundamentals

Q1

```
Example.java x
1  import java.util.*;
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5
6          System.out.print("Input First number : ");
7          int first_no = input.nextInt();
8
9          System.out.print("Input Second number : ");
10         int second_no = input.nextInt();
11
12         System.out.print('\n');
13
14
15         if (first_no > second_no){
16             int sum = first_no + second_no;
17
18             System.out.println(first_no + " + " + second_no + " = " + sum);
19
20         }else{
21             System.out.println( second_no + " is greater than the " + first_no );
22
23         }
24
25     }
26 }
27
```

```
C:\WINDOWS\SYSTEM32\cmd x + v
Input First number : 12
Input Second number : 10

12 + 10 = 22

-----
(program exited with code: 0)
Press any key to continue . . .
```

```
C:\WINDOWS\SYSTEM32\cmd x + v
Input First number : 10
Input Second number : 12

12 is greater than the 10

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q2

```
Example.java x
1  import java.util.*;
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Input integer number : ");
6          int x = input.nextInt();
7
8          if ( x < 0){
9              int num = x * (-1);
10             System.out.println(" Absolute number of "+x+" is " + num );
11         }else{
12             System.out.println(" Absolute number of "+x+" is "+ x );
13         }
14     }
15 }
16 }
17 }
```

```
C:\WINDOWS\SYSTEM32\cmd x + v
Input integer number : -50
Absolute number of -50 is 50

-----
(program exited with code: 0)
Press any key to continue . . .
```

```
C:\WINDOWS\SYSTEM32\cmd x + v
Input integer number : 50
Absolute number of 50 is 50

-----
(program exited with code: 0)
Press any key to continue . . .
```

Q3

```
Example.java x
1  import java.util.*;
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Input Chemistry Marks : ");
6          int chem_marks = input.nextInt();
7
8          System.out.print("Input Physics Marks : ");
9          int phy_marks = input.nextInt();
10
11         System.out.print("Input Combined Maths Marks : ");
12         int com_marks = input.nextInt();
13
14         System.out.println('\n');
15
16         int total_marks = chem_marks + phy_marks + com_marks;
17         int avg_marks = total_marks /3;
18
19
20         if(avg_marks > 75){
21             System.out.println("Pass");
22         }else{
23             System.out.println("Fail");
24         }
25
26
27
28     }
29 }
30
```

```
C:\WINDOWS\SYSTEM32\cmd x + v
Input Chemistry Marks : 90
Input Physics Marks : 85
Input Combined Maths Marks : 78

Pass
-----
```

```
C:\WINDOWS\SYSTEM32\cmd x + v
Input Chemistry Marks : 50
Input Physics Marks : 45
Input Combined Maths Marks : 60

Fail
-----
```

Q4

```
Example.java x
1  import java.util.*;
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Input Unit Price : ");
6          double unit_price = input.nextDouble();
7
8          System.out.print("Input Amount : ");
9          double amount = input.nextDouble();
10
11         System.out.println('\n');
12
13         double total = unit_price * amount;
14
15         if(total >= 1500){
16             System.out.println("You are entitled to the super draw. ");
17
18         }else{
19             System.out.println("Try again");
20         }
21     }
22 }
23
24
```

```
C:\WINDOWS\SYSTEM32\cmd x + v
Input Unit Price : 250.00
Input Amount : 10

You are entitled to the super draw.
-----
```

```
C:\WINDOWS\SYSTEM32\cmd x
Input Unit Price : 500.00
Input Amount : 2

Try again
-----
```

Q5

```
Example.java x
1  import java.util.*;
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Input Unit Price : ");
6          double unit_price = input.nextDouble();
7
8          System.out.print("Input Amount : ");
9          double amount = input.nextDouble();
10
11         System.out.println('\n');
12
13         double total = unit_price * amount;
14
15         if(total >= 500){
16             double discount = total * 0.05;
17             double new_total = total + discount;
18             System.out.println("Discount : " + discount);
19             System.out.println("Total : " + new_total);
20         }else{
21             System.out.println("No discount given");
22         }
23     }
24 }
25
26
```

```
C:\WINDOWS\SYSTEM32\cmd x +
Input Unit Price : 1000.00
Input Amount : 2

Discount : 100.0
Total : 2100.0
-----
```

```
C:\WINDOWS\SYSTEM32\cmd x -
Input Unit Price : 200.00
Input Amount : 2

No discount given
-----
```

Q6

```
Example.java x
1  import java.util.*;
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Input Year : ");
6          int year = input.nextInt();
7
8          System.out.println('\n');
9
10         int leap_year = year / 4;
11         int leap = year % 4;
12
13         if(leap == 0){
14             System.out.println(year + " is leap Year. ");
15         }else{
16             System.out.println(year + " is not leap Year. ");
17         }
18     }
19 }
20 }
21 }
```

```
C:\WINDOWS\SYSTEM32\cmd x
Input Year : 2020

2020 is leap Year.
-----
```

```
C:\WINDOWS\SYSTEM32\cmd x
Input Year : 2026

2026 is not leap Year.
-----
```

Q7

```
Example.java x
1  import java.util.*;
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Input First Number : ");
6          int first_no = input.nextInt();
7
8          System.out.print("Input Second Number : ");
9          int second_no = input.nextInt();
10
11         System.out.print("Input Third Number : ");
12         int third_no = input.nextInt();
13
14         System.out.println('\n');
15
16         if(first_no > second_no & first_no > third_no){
17             System.out.println(" Maximum number is : " + first_no);
18         }else if (second_no > third_no){
19             System.out.println(" Maximum number is : " + second_no);
20         }else{
21             System.out.println(" Maximum number is : " + third_no);
22         }
23     }
24 }
25
```

```
C:\WINDOWS\SYSTEM32\cmd x
Input First Number : 12
Input Second Number : 45
Input Third Number : 10

Maximum number is : 45
_____
```

```
Input First Number : 78
Input Second Number : 45
Input Third Number : 12
```

```
Maximum number is : 78
_____
```

Q8

☐ class Example{
 public static void main(String[] args) {
 int x = 10;
 if (x == 10){
 }
 }
}

☐ class Example{
 public static void main(String[] args) {
 int x = 10;
 if((x=100)!=10){
 }
 }
}

☐ class Example{
 public static void main(String[] args) {
 int x = 10;
 if((x=100)>0==true){
 }
 }
}

Q9

```
class Example{  
    public static void main(String[] args) {  
        int x = 10;  
        boolean b=true;  
        if(b){  
            }  
        }  
    }  
}
```

```
class Example{  
    public static void main(String[] args) {  
        int x = 10;  
        boolean b=true;  
        if(b=false){  
            }  
        }  
    }  
}
```

```
class Example{  
    public static void main(String[] args) {  
        int x = 10;  
        boolean b=true;  
        if(b==false){  
            }  
        }  
    }  
}
```

Q10

```
Example.java ✕
1  import java.util.*;
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Input an integer number : ");
6          int x = input.nextInt();
7          System.out.println('\n');
8          int no;
9
10         no = x /2;
11         no = x % 2;
12         if (no == 0){
13             System.out.println("The number is even");
14         }else{
15             System.out.println("The number is odd");
16         }
17
18     }
19 }
20
```

Input an integer number : 24

The number is even

Input an integer number : 65

The number is odd

Q11

- A. `System.out.println(b); //true`
- B. `System.out.println(b=true); //true`
- C. `System.out.println(b==true); //true`
- D. `System.out.println(b!=true); //false`
- E. `System.out.println(b=true==true); //true`
- F. `System.out.println((b=true)==false); //false`
- G. `System.out.println(b=(true!=false)); //true`

Q12

- A. `System.out.println(x=9); //9`
- B. `System.out.println(x==9); //true`
- C. `System.out.println(x=9!=10); //error`
- D. `System.out.println((x=9)==10); //false`
- E. `System.out.println((x=9)<=10); //true`

Q13

```
int x=20,y=60;
boolean bool;
System.out.println(x=10); //10
System.out.println(bool=true);//true
System.out.println(x=10>0); //error
System.out.println((x=10)>0);//true
System.out.println(bool=(x=10)>0);//true
System.out.println(bool=x+y>100);//false
```

Q14

```
System.out.println(b+s+x+""+f+d+c+bool);//2351.521.231ctrue
System.out.println(""+b+s+x+f+d+c+bool); //101001251.521.231ctrue
System.out.println(b+s+x+f+d+c+""+bool); //356.731true
System.out.println(b+s+x+f+d+c+bool+""); //error
System.out.println(bool+b+f+d+c+""+x+l); //error
```

Q15

```
System.out.println( i ==1); //true
```

```
System.out.println( j ==3); //false
```

```
System.out.println( ( i >=1) && ( j <4) ); //true
```

```
System.out.println( ( m <=99) & ( k < m ) ); //false
```

```
System.out.println( ( j >= i ) || ( k == m ) ); //true
```

```
System.out.println( ( k + m < j ) || (3- j >= k)); // false
```

```
System.out.println( !( k > m ) ); //false
```

Q16

```
int x=1; // 1 2 3
```

```
int x=2; // 2 3
```

```
int x=3; // 3
```

```
int x=4; // 4 1 2 3
```

```
int x=0; // 4 1 2 3
```

```
int x=5; // 4 1 2 3
```

Q17

```
class Example{  
    public static void main(String args[]){  
        int a=-5;  
        int b=-2;  
        a%=b;  
        a/=b;  
        b=a>0?0:a;  
        System.out.println(a+" "+b); //0 0  
    }  
}
```

Q18

Line 7

Line 9

Line 10

Q19

`int x=1;//1`

`int x=2;//2 3 1`

`int x=3;//3 1`

`int x=4;//Wrong`

`int x=0;// Wrong`

`int x=5;// Wrong`

Q20

```
Example.java ✕
1  import java.util.*;
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Input an integer : ");
6          int x = input.nextInt();
7
8          System.out.println('\n');
9
10         if( x < 0){
11             System.out.println("Negative integer");
12         }else if(x > 0){
13             System.out.println("Positive integer");
14         }else if(x==0){
15             System.out.println("Zero");
16         }
17
18
19     }
20 }
21
```

```
C:\WINDOWS\SYSTEM32\cmd
Input an integer : -20

Negative integer

-----
```

```
C:\WINDOWS\SYSTEM32\cmd ✕
Input an integer : 50

Positive integer

-----
```

Q21

```
Example.java ✕
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Input First Number : ");
6          int first_no = input.nextInt();
7
8          System.out.print("Input Second Number : ");
9          int second_no = input.nextInt();
10
11         System.out.println('\n');
12         int x,y,dif;
13
14         if(first_no < 0 & second_no < 0){
15             x = first_no * (-1);
16             y = second_no * (-1);
17             dif = x - y;
18             System.out.println(dif);
19         }else if(first_no > 0 & second_no > 0){
20             dif = first_no - second_no;
21             System.out.println(dif);
22         }else if(first_no > 0 & second_no < 0){
23             y = second_no * (-1);
24             dif = first_no - y;
25             System.out.println(dif);
26         }else{
27             x = first_no * (-1);
28             dif = x - second_no;
29             System.out.println(dif);
30         }
31
32     }
33 }
34
```

```
C:\WINDOWS\SYSTEM32\cmd - X
Input First Number : -10
Input Second Number : -5

5
-----
```

```
C:\WINDOWS\SYSTEM32\cmd - X
Input First Number : -15
Input Second Number : 5

10
-----
```

Q22

char x='A';

int x=65;

int x=65536;

byte x=65;

short x=66 ;

Q23

```
Example.java x
1  import java.util.*;
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Input First Number : ");
6          int first_no = input.nextInt();
7
8          System.out.print("Input Second Number : ");
9          int second_no = input.nextInt();
10
11         if(first_no > second_no){
12             System.out.println("The first number is greater than the second number");
13         }else if(first_no < second_no){
14             System.out.println("The first number is less than the second number.");
15         }else{
16             System.out.println("Both are equal.");
17         }
18     }
19 }
20
```

```
C:\WINDOWS\SYSTEM32\cmd  x  +  v
Input First Number : 25
Input Second Number : 39
The first number is less than the second number.
-----
```

```
C:\WINDOWS\SYSTEM32\cmd  x  +  v
Input First Number : 39
Input Second Number : 25
The first number is greater than the second number
-----
```

Q24

```
Example.java x
1  import java.util.*;
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Input First Number : ");
6          int first_no = input.nextInt();
7
8          System.out.print("Input Second Number : ");
9          int second_no = input.nextInt();
10
11         System.out.print("Input Third Number : ");
12         int third_no = input.nextInt();
13
14         int n1,n2,n3;
15
16         n1 = first_no % 10;
17         n2 = second_no % 10;
18         n3 = third_no % 10;
19
20         if(n1 == n2 | n2 == n3 | n1 == n3)
21             System.out.println("True");
22
23
24
25     }
26 }
27
```

```
C:\WINDOWS\SYSTEM32\cmd x +
Input First Number : 123
Input Second Number : 453
Input Third Number : 458
True
-----
```

Q25

```
Example.java x
1 import java.util.*;
2 class Example{
3     public static void main(String args[]){
4         Scanner input = new Scanner(System.in);
5         System.out.print("Input First Number : ");
6         int first_no = input.nextInt();
7
8         System.out.print("Input Second Number : ");
9         int second_no = input.nextInt();
10
11        System.out.print("Input Third Number : ");
12        int third_no = input.nextInt();
13
14        int n1,n2,n3;
15
16        n1 = first_no - second_no;
17        n2 = first_no - third_no;
18        n3 = second_no - third_no;
19
20        if(first_no > n1 || first_no > n2 || first_no > n3 || second_no > n1 || second_no > n2 || second_no > n3 || third_no > n1 || third_no > n2 || third_no > n3){
21            System.out.println("True");
22        }else{
23            System.out.println("False");
24        }
25
26    }
27 }
28 }
```

```
C:\WINDOWS\SYSTEM32\cmd
Input First Number : 14
Input Second Number : 3
Input Third Number : 78
True
-----
```

Q26

```
Example.java x
1  import java.util.*;
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Input Selling Price : ");
6          int sell = input.nextInt();
7
8          System.out.print("Input Cost of the Product : ");
9          int cost = input.nextInt();
10
11         System.out.println('\n');
12
13         if(sell > cost){
14             int Profit = sell - cost;
15             System.out.println("Profit");
16         }else{
17             int Loss = cost - sell;
18             System.out.println("Loss");
19         }
20
21     }
22 }
23
24
```

```
Input Selling Price : 2500
Input Cost of the Product : 2000
```

Profit

```
Input Selling Price : 1000
Input Cost of the Product : 1200
```

Loss

Q27

```
Example.java x
1  import java.util.*;
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Input First Number : ");
6          int first_no = input.nextInt();
7
8          System.out.print("Input Second Number : ");
9          int second_no = input.nextInt();
10
11         System.out.print("Input Third Number : ");
12         int third_no = input.nextInt();
13
14         System.out.println('\n');
15
16         if(first_no > second_no & second_no > third_no){
17             System.out.print("Decreasing");
18         }else if(third_no > second_no & second_no > first_no){
19             System.out.print("Increasing");
20         }else{
21             System.out.print("Neither increasing nor decreasing order");
22         }
23
24     }
25 }
26 }
27 }
```

```
Input First Number : 78
Input Second Number : 45
Input Third Number : 12

Decreasing
-----
```

```
C:\WINDOWS\SYSTEM32\cmd.exe
Input First Number : 54
Input Second Number : 78
Input Third Number : 45

Neither increasing nor decreasing order
-----
```

Q28

```
Example.java x
1  import java.util.*;
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Input Age  : ");
6          int age = input.nextInt();
7
8          System.out.print("Input Weigh  : ");
9          int weigh = input.nextInt();
10
11         System.out.println('\n');
12
13         if(weigh >=50 && age >=18){
14             System.out.println("Eligible for blood donation. ");
15         }else{
16             System.out.println("Ineligible for blood donation. ");
17         }
18
19     }
20 }
21
22
```

```
Input Age  : 19
Input Weigh  : 60

Eligible for blood donation.
-----
```

```
Input Age  : 17
Input Weigh  : 58

Ineligible for blood donation.
-----
```

Q29

```
Example.java ✕
1  import java.util.*;
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Input First Number : ");
6          int first_no = input.nextInt();
7
8          System.out.print("Input Second Number : ");
9          int second_no = input.nextInt();
10
11         System.out.println('\n');
12
13         if(first_no >0 & second_no >0 || first_no <0 & second_no <0 ){
14             System.out.println("True");
15         }
16
17     }
18 }
19
```

```
Input First Number : 20
Input Second Number : 14
```

True

```
Input First Number : -20
Input Second Number : -15
```

True

Q30

```
Example.java X
1  import java.util.*;
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Input Character : ");
6          char
7          | ch = input.next().charAt(0);
8
9          System.out.println('\n');
10
11         int value = ch;
12
13         if(value >= 65 && value <=90){
14             System.out.println(ch+" is a uppercase Character. ");
15         }else if(value >=97 && value <=122){
16             System.out.println(ch+" is a lowercase Character. ");
17         }else{
18             System.out.println(ch+" is a neighter uppercase or lowercase Character. ");
19         }
20     }
21 }
22
```

```
C:\WINDOWS\SYSTEM32\cmd X
Input Character : C

C is a uppercase Character.

-----
```

```
Input Character : %

% is a neighter uppercase or lowercase Character.

-----
```

Q31

```
Example.java x
1  import java.util.*;
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Input Number : ");
6          int no = input.nextInt();
7
8          System.out.println('\n');
9
10         int n1 = no % 10;
11         int n2 = no % 7;
12
13         if(n1 == 7 || n2 == 0){
14             System.out.println(no + " is a Buzz Number.");
15         }else{
16             System.out.println(no + " is not a Buzz Number.");
17         }
18     }
19 }
20 }
21 }
```

```
Input Number : 70

70 is a Buzz Number.

-----
```

```
C:\WINDOWS\SYSTEM32\cmd x
Input Number : 458

458 is not a Buzz Number.

-----
```

Q32

```
Example.java ✕
1  import java.util.*;
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Input number of classes held : ");
6          int held = input.nextInt();
7
8          System.out.print("Input number of classes attended : ");
9          int attended = input.nextInt();
10
11         System.out.print("Input do you have a medical cause('Yes' or 'No') : ");
12         char medical = input.next().charAt(0);
13         System.out.println('\n');
14
15         double per = (attended/held)*100;
16
17         if(per >=70||medical == 'Y'){
18             System.out.println(" A student will be allowed to sit for the exam.");
19         }else{
20             System.out.println("A student will not be allowed to sit for the exam.");
21         }
22     }
23 }
24 }
25 }
```

```
Input number of classes held : 50
Input number of classes attended : 10
Input do you have a medical cause('Yes' or 'No') : Yes
```

```
A student will be allowed to sit for the exam.
```

```
Input number of classes held : 50
Input number of classes attended : 10
Input do you have a medical cause('Yes' or 'No') : Yes
```

```
A student will be allowed to sit for the exam.
```

Q33

```
Example.java ✕
1  import java.util.*;
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Input salary : ");
6          double salary = input.nextDouble();
7
8          System.out.print("Input year of service : ");
9          int service = input.nextInt();
10
11         if(service < 5){
12             double bonus = salary * 0.10;
13             double total_salary = salary +(salary * 0.10);
14             System.out.println("Bonus : " + bonus);
15             System.out.println("Total salary : " +total_salary);
16         }else if(service < 10){
17             double bonus = salary * 0.15;
18             double total_salary = salary +(salary * 0.15);
19             System.out.println("Bonus : " + bonus);
20             System.out.println("Total salary : " +total_salary);
21         }else{
22             double bonus = salary * 0.25;
23             double total_salary = salary + (salary * 0.25);
24             System.out.println("Bonus : " + bonus);
25             System.out.println("Total salary : " +total_salary);
26         }
27     }
28 }
29
30
```

```
C:\WINDOWS\SYSTEM32\cmd
Input salary : 50000
Input year of service : 20
Bonus : 12500.0
Total salary : 62500.0
-----
```

Q34

```
Example.java x
1  import java.util.*;
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Input number of books : ");
6          int books = input.nextInt();
7
8          double cost = 100 * books;
9          double discount = 0, total;
10         if(cost > 5000){
11             discount = cost * 0.10;
12         }
13         total = cost - discount;
14         System.out.println("Subtotal : "+cost );
15
16         if(discount == 0){
17             System.out.println("Discount : - " );
18         }else{
19             System.out.println("Discount : " +discount);
20         }
21         System.out.println("Total : " +total);
22     }
23 }
24
```

```
C:\WINDOWS\SYSTEM32\cmd x
Input number of books : 25
Subtotal : 2500.0
Discount : -
Total : 2500.0
-----
```

```
C:\WINDOWS\SYSTEM32\cmd x
Input number of books : 60
Subtotal : 6000.0
Discount : 600.0
Total : 5400.0
-----
```

Q35

```
Example.java ✕
1  import java.util.*;
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Input Temperature : ");
6          int temp = input.nextInt();
7
8          String act;
9
10         act = temp >=80 ? "Swimming":(temp>=60 &&temp<80 ? "Tennis":(temp>=40 && temp<60 ? "Golf":"Skiing"));
11         System.out.println(act);
12
13     }
14 }
15
```

Q36

```
Example.java ✕
1  import java.util.*;
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Enter the Character : ");
6          char character = input.next().charAt(0);
7
8          int ch = character;
9
10         System.out.print('\n');
11
12         if(ch ==97 || ch ==101 || ch==105 || ch==111 || ch==117 || ch==65 || ch==69 || ch==73 || ch==79 || ch==85){
13             System.out.print(character+" is a vowel.");
14         }else{
15             System.out.print(character+" is a consonant.");
16         }
17     }
18 }
19
```

```
Enter the Character : a
a is a vowel.
```

Q37

```
Example.java x
1  import java.util.*;
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Enter Salary : ");
6          int salary = input.nextInt();
7
8          System.out.print("Enter number indicating the performance rating (1=excellent, 2=good, and 3=poor) : ");
9          int rate = input.nextInt();
10
11         double new_salary;
12
13         System.out.print('\n');
14
15         if(rate ==1){
16             new_salary = salary + (salary * 0.06);
17             System.out.println("New Salary :"+new_salary);
18
19         }else if(rate ==2){
20             new_salary = salary + (salary * 0.04);
21             System.out.println("New Salary :"+new_salary);
22
23         }else if(rate==3){
24             new_salary = salary + (salary * 0.015);
25             System.out.println("New Salary :"+new_salary);
26         }
27     }
28 }
29
```

Q38

```
Example.java x
1  import java.util.*;
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Input Average Marks : ");
6          double avg = input.nextDouble();
7
8          System.out.print("Input Attendance : ");
9          int attendance = input.nextInt();
10
11         if(avg >= 50 & attendance >= 80){
12             System.out.println("You are eligible to sit for the O/L examination.");
13         }
14     }
15 }
16
```

```
Input Average Marks : 60
Input Attendance : 80
You are eligible to sit for the O/L examination.
-----
```

Q39

```
Example.java x
1  import java.util.*;
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Input time(HH:MM) : ");
6          String inputTime = input.nextLine();
7
8          String time = inputTime.split(":") [0];
9
10         int hours = Integer.parseInt(time);
11
12         if(hours>=0 && hours<12){
13             System.out.println("Good morning!");
14         }else if(hours>=12 && hours < 16){
15             System.out.println("Good afternoon!");
16         }else if(hours>=16 && hours <19){
17             System.out.println("Good evening!");
18         }else{
19             System.out.println("Good night!");
20         }
21     }
22 }
23
```

```
C:\WINDOWS\SYSTEM32\cmd
Input time(HH:MM) : 13:45
Good afternoon!
_____
```

Q40

```
Example.java ✕
1  import java.util.*;
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Enter Year : ");
6          int year = input.nextInt();
7
8          System.out.print("Enter month : ");
9          String month = input.next();
10
11         switch(month){
12             case "january":
13             case "march":
14             case "may":
15             case "july":
16             case "august":
17             case "octomber":
18             case "december":
19                 System.out.println("31 Days");
20                 break;
21             case "april":
22             case "june":
23             case "september":
24             case "november":
25                 System.out.println("30 Days");
26                 break;
27             default:
28                 System.out.println("28 Days");
29         }
30         int leap_year = year % 4;
31         if(leap_year==0){
32             System.out.println(year + " is a leap year.");
33         }else{
34             System.out.println(year + " is not a leap year.");
35         }
36     }
37 }
38
```

```
C:\WINDOWS\SYSTEM32\cmd
Enter Year : 2015
Enter month : december
31 Days
2015 is not a leap year.
-----
```

Q41

```
Example.java x
1  import java.util.*;
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Enter number of copies : ");
6          int copy = input.nextInt();
7          int total;
8
9          System.out.print('\n');
10
11         if(copy < 99){
12             total = copy * 30;
13             System.out.println("Total price : "+ total);
14             System.out.println(" Per copy : Rs.30.00");
15
16         }else if( copy > 100 && copy < 499){
17             total = copy * 28;
18             System.out.println("Total price : "+ total);
19             System.out.println(" Per copy : Rs.28.00");
20
21         }else if(copy > 500 && copy < 799){
22             total = copy * 27;
23             System.out.println("Total price : "+ total);
24             System.out.println(" Per copy : Rs.27.00");
25
26         }else if(copy > 800 && copy < 1000){
27             total = copy * 26;
28             System.out.println("Total price : "+ total);
29             System.out.println(" Per copy : Rs.26.00");
30
31         }else if(copy > 1000 ){
32             total = copy * 25;
33             System.out.println("Total price : "+ total);
34             System.out.println(" Per copy : Rs.25.00");
35         }
36     }
37 }
```

Q42

```
Example.java x
1  import java.util.*;
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Input Inches : ");
6          int x = input.nextInt();
7
8          System.out.println('\n');
9          String size;
10         size = x==28 || x==29 ? "X-small":(x==30 || x==31 ? "Small":(x==32 || x<=34?"Medium":(x==36 || x<=38 ?"Large":"X-Large")));
11         System.out.println(size);
12
13     }
14 }
```

Q43

```
Example.java x
1  import java.util.*;
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Enter the First Number : ");
6          int first_no = input.nextInt();
7
8          System.out.print("Enter the Second Number : ");
9          int second_no = input.nextInt();
10
11         System.out.print("Enter the Operator(+,-,*,/,%,^) : ");
12         char operator = input.next().charAt(0);
13
14         System.out.print('\n');
15
16         int addition, subtraction, multiplication, division, remainder, power;
17
18         if(operator=='+'){
19             addition = first_no + second_no;
20             System.out.println(first_no + " + " + second_no + " = " + addition);
21
22         }else if(operator=='-'){
23             subtraction = first_no - second_no;
24             System.out.println(first_no + " - " + second_no + " = " + subtraction);
25
26         }else if(operator=='*'){
27             multiplication = first_no * second_no;
28             System.out.println(first_no + " * " + second_no + " = " + multiplication);
29
30         }else if(operator=='/'){
31             division = first_no / second_no;
32             System.out.println(first_no + " / " + second_no + " = " + division);
33
34         }else if(operator=='%'){
35             remainder = first_no % second_no;
36             System.out.println(first_no + " % " + second_no + " = " + remainder);
37
38         }else if(operator=='^'){
39             power = (int) Math.pow(first_no, second_no);
40             System.out.println(first_no + " ^ " + second_no + " = " + power);
41
42         }
43     }
44 }
45
```

Q44

```
Example.java x
1  import java.util.*;
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Input Date : ");
6          int date = input.nextInt();
7
8          switch(date){
9              case 6:
10             case 13:
11             case 20:
12             case 27:
13                 System.out.println("Sunday");
14                 break;
15             case 7:
16             case 14:
17             case 21:
18             case 28:
19                 System.out.println("Monday");
20                 break;
21             case 1:
22             case 8:
23             case 15:
24             case 22:
25                 System.out.println("Tuesday");
26                 break;
27             case 2:
28             case 9:
29             case 16:
30             case 23:
31                 System.out.println("Wednesday");
32                 break;
33             case 3:
34             case 10:
35             case 17:
36             case 24:
37                 System.out.println("Thursday");
38                 break;
39             case 4:
40             case 11:
41             case 18:
42             case 25:
43                 System.out.println("Friday");
44                 break;
45             default:
46                 System.out.println("Saturday");
47             }
48         }
49     }
```

Q45

```
Example.java ✕
1  import java.util.*;
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Input Row Number : ");
6          int row = input.nextInt();
7
8          System.out.print("Input Column Number : ");
9          int column = input.nextInt();
10
11         int r1 = row % 2;
12         int c1 = column % 2;
13
14
15         if(r1==0 && c1==0){
16             System.out.println("White");
17         }else if(r1==1 && c1==1){
18             System.out.println("White");
19         }else{
20             System.out.println("Black");
21         }
22     }
23 }
24
```

Q46

```
Example.java x
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Enter First number  : ");
6          int first_no = input.nextInt();
7
8          System.out.print("Enter Second number  : ");
9          int second_no = input.nextInt();
10
11         System.out.print("Enter Third number  : ");
12         int third_no = input.nextInt();
13
14         System.out.print('\n');
15
16         int max;
17         double x,y;
18         if(first_no > second_no && first_no > third_no){
19             max = first_no;
20         }else if(second_no > third_no){
21             max = second_no;
22         }else{
23             max = third_no;
24         }
25
26         if(max==first_no){
27             x = second_no;
28             y = third_no;
29         }else if(max==second_no){
30             x = first_no;
31             y = third_no;
32         }else{
33             x = first_no;
34             y = second_no;
35         }
36
37         double p=(x*x) + (y*y);
38         double m=(max*max);
39         if(p==m){
40             System.out.println("true");
41         }else{
42             System.out.println("flase");
43         }
44     }
45 }
46
47
48 }
```

Q47

```
Example.java x
1  import java.util.*;
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Enter Month   : ");
6          int month = input.nextInt();
7
8          System.out.print("Enter Day    : ");
9          int day = input.nextInt();
10
11         if(month<=2||(month==12 && day==21)||(month==3 && day==19)){
12             System.out.println("winter");
13
14         }else if( month>3 && month<=5 ||(month==3 && day==20)||(month==6 && day==20)){
15             System.out.println("Spring");
16
17         }else if(month==6 && day==21||(month==9 && day==21)||(month>6 && month<9)){
18             System.out.println("Summer");
19
20         }else if(month>9 && month < 12||(month==9 && day==22)||(month==12 && day==20)){
21             System.out.println("Autumn");
22         }
23     }
24 }
25 }
```

Q48

```
Example.java x
3 public static void main(String args[]){
4     Scanner input = new Scanner(System.in);
5     System.out.print("Enter Month : ");
6     int month = input.nextInt();
7
8     System.out.print("Enter Day : ");
9     int day = input.nextInt();
10
11     if(month==12 && day>=22||(month<2 && day <20 )){
12         System.out.println("Capricornus");
13
14     }else if(month==1 && day >=20||(month==2 && day<19)){
15         System.out.println("Aquarius");
16
17     }else if(month==2 && day> 18||(month==3 && day<21)){
18         System.out.println("Pisces");
19
20     }else if(month==3 && day >=21||(month==4 && day<20)){
21         System.out.println("Aries");
22
23     }else if(month==4 && day >=20||(month==5 && day<21)){
24         System.out.println("Taurus");
25
26     }else if(month==5 && day >=21||(month==6 && day<22)){
27         System.out.println("Gemini");
28
29     }else if(month==6 && day >=22||(month==7 && day<23)){
30         System.out.println("Cancer");
31
32     }else if(month==7 && day >=23||(month==8 && day<23)){
33         System.out.println("Leo");
34
35     }else if(month==8 && day >=23||(month==9 && day<23)){
36         System.out.println("Virgo");
37
38     }else if(month==9 && day >=23||(month==10 && day<24)){
39         System.out.println("Libra");
40
41     }else if(month==10 && day >=24||(month==11 && day<22)){
42         System.out.println("Scorpius");
43
44     }else if(month==11 && day >=22||(month==12 && day<22)){
45         System.out.println("Sagittarius");
46     }
47 }
48
49 }
```

Q49

```
Example.java x
1  import java.util.*;
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Enter Basic Salary : ");
6          int salary = input.nextInt();
7
8          System.out.print('\n');
9
10         double x,y;
11
12         if(salary<=10000){
13             x = salary * 0.20 + salary * 0.60;
14             y = salary + x;
15         }else if(salary <=20000){
16             x = salary * 0.25 + salary * 0.70;
17             y = salary + x;
18         }else{
19             x = salary * 0.30 + salary * 0.75;
20             y = salary + x;
21         }
22
23         System.out.println("Total Salary : " + y);
24
25     }
26 }
```

```
C:\WINDOWS\SYSTEM32\cmd
Enter Basic Salary : 30000

Total Salary : 61500.0

-----
```

Q50

```
Example.java ✕
1  import java.util.*;
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Input first angle : ");
6          int first = input.nextInt();
7
8          System.out.print("Input second angle : ");
9          int second = input.nextInt();
10
11         System.out.print("Input third angle : ");
12         int third = input.nextInt();
13
14         System.out.print('\n');
15
16         int x;
17
18         x= first + second + third;
19
20         if(x==180){
21             System.out.println("A triangle can be formed.");
22         }else{
23             System.out.println("A triangle can not be formed.");
24         }
25     }
26 }
```

Q51

```
Example.java ✕
1  import java.util.*;
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Input Age : ");
6          int age = input.nextInt();
7
8          if(age >65){
9              System.out.println("Senior");
10         }else if(age <=65 && age >20){
11             System.out.println("Adult");
12
13         }else if(age <=20 && age >13){
14             System.out.println("Teenager");
15         }else if(age >=13 && age >1){
16             System.out.println("Child");
17         }else{
18             System.out.println("Infant");
19         }
20
21     }
22 }
23
24
```

Q52

```
Example.java x
1  import java.util.*;
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Enter X Quadrant : ");
6          int x = input.nextInt();
7
8          System.out.print("Enter Y Quadrant : ");
9          int y = input.nextInt();
10
11         if(x>0 && y>0){
12             System.out.println("Quadrant 1");
13         }else if(x<0 && y<0){
14             System.out.println("Quadrant 3");
15         }else if(x<0 && y>0){
16             System.out.println("Quadrant 2");
17         }else{
18             System.out.println("Quadrant 4");
19         }
20     }
21 }
```

```
Enter X Quadrant : -5
Enter Y Quadrant : 2
Quadrant 2
-----
```

Q53

```
Example.java ✕
1  import java.util.*;
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Input SBP Value : ");
6          int sbp = input.nextInt();
7
8          System.out.print("Input DBP Value : ");
9          int dbp = input.nextInt();
10
11         if(sbp >=130 && dbp>=90){
12             System.out.println("High Pressure");
13         }else if(sbp <=100 && dbp<=70){
14             System.out.println("Low Pressure");
15         }else if((sbp >=100 && sbp <131) && (dbp>=70 && dbp <91)){
16             System.out.println("Normal ");
17         }
18     }
19 }
20
```

Q58

```
Example.java x
1  import java.util.*;
2  class Example{
3      public static void main(String args[]){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Input Year : ");
6          int y = input.nextInt();
7
8          System.out.print("Input Month : ");
9          int m = input.nextInt();
10
11         System.out.print("Input Day : ");
12         int d = input.nextInt();
13
14         double y0,d0,m0,x;
15
16         int a = (14 - m ) / 12;
17         y0 = y - a;
18         x = y0 + y0 /4 - y0/100 + y0/400;
19         m0 = m + 12 * a -2;
20         double b = (31 * m0) /12;
21         d0 =(int) (d + x + b) % 7;
22
23         if(d0==0){
24             System.out.println("Day of the week : Sunday");
25         }else if(d0==1){
26             System.out.println("Day of the week : Monday");
27         }else if(d0==2){
28             System.out.println("Day of the week : Tuesday");
29         }else if(d0==3){
30             System.out.println("Day of the week : Wednesday");
31         }else if(d0==4){
32             System.out.println("Day of the week : Thursday");
33         }else if(d0==5){
34             System.out.println("Day of the week : Friday");
35         }else if(d0==6){
36             System.out.println("Day of the week : Saturday");
37         }
38
39     }
40 }
```

